Lecture 10: Floating Rate Notes and Interest Rate Swaps

Questions

These five multiple-choice questions follow the lecture's reasoning with different numbers. Work through the problem before reading the solutions. Keep full precision until the final answer.

Use the SOFR rate set at the semester's start. All rates are annual; payments are semiannual. Net cash flow always means received minus paid.

1. Fix the first FRN coupon

An FRN has $2,000 face value, a coupon of SOFR + 2%, and semiannual payments. SOFR is 4% at t=0 and 5% at t=0.5. What coupon is paid at t=0.5 under the lecture's start-of-period reset convention?

A. $60

B. $40

C. $120

D. $70

2. Follow the FRN to maturity

The same FRN matures at t=2. SOFR observations at t=0, 0.5, 1, 1.5 and 2 are 4%, 5%, 3%, 2% and 6%. What is the final total bond payment at t=2?

A. $40

B. $2,080

C. $2,060

D. $2,040

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3. Calculate a swap net receipt

Firm A receives 6% fixed and pays SOFR on a $2,000 swap notional. The semester's SOFR is 4%. What is A's net cash flow at the end of the semester?

A. +$20

B. +$60

C. -$20

D. +$40

4. Add four swap cash flows

Keep the receive-6%-fixed/pay-SOFR swap on $2,000. SOFR used for its four semesters is 4%, 5%, 3% and 2%. What is A's total net cash flow?

A. +$240

B. +$140

C. +$100

D. -$100

5. Combine fixed debt and a swap

A company has fixed-rate debt at 6%. On a matching notional it receives 6% fixed and pays SOFR through an IRS. What is its final annual borrowing rate?

A. SOFR + 6%

B. SOFR

C. 6%

D. 6% - SOFR

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Worked solutions

1. Fix the first FRN coupon - Answer A

Step 1. Read the payment date. The payment at t=0.5 uses SOFR fixed at t=0, which is 4%.

Step 2. Add the spread. The annual coupon is 4% + 2% = 6%.

Step 3. Convert to one semester. 2,000 x 6% / 2 = $60.

$$C_{0.5} = 2000(0.04 + 0.02)/2 = 60$$

Interpretation. The new 5% rate observed on payment day sets the next coupon.

Check your reasoning. Do not use a newly observed rate for the semester that has just ended.

2. Follow the FRN to maturity - Answer D

Step 1. Select the applicable reset. The last coupon uses 2% observed at t=1.5; the 6% at maturity is unused.

Step 2. Calculate the last coupon. 2,000 x (2% + 2%) / 2 = $40.

Step 3. Include the bond principal. The bond repays $2,000 plus its final $40 coupon.

$$CF_2 = 2000 + 2000(0.02 + 0.02)/2 = 2040$$

Interpretation. Bond principal is repaid at maturity. That is separate from a swap's notional.

Check your reasoning. Do not omit principal when asked for the total bond payment.

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3. Calculate a swap net receipt - Answer A

Step 1. Calculate the fixed receipt. 2,000 x 6% / 2 = $60.

Step 2. Calculate the floating payment. 2,000 x 4% / 2 = $40.

Step 3. Subtract payments from receipts. Net cash flow = 60 - 40 = +$20.

$$NCF = 2000(0.06 - 0.04)/2 = 20$$

Interpretation. A receives $20; its counterparty pays $20.

Check your reasoning. Net cash flow is received minus paid, not paid minus received.

4. Add four swap cash flows - Answer C

Step 1. Keep the fixed leg unchanged. Each receipt is $60, for $240 in total.

Step 2. Calculate the floating leg. Payments are $40, $50, $30 and $20, totaling $140.

Step 3. Add net receipts. The semester nets are +20, +10, +30 and +40; total +$100.

$$\sum NCF = 4(60) - (40 + 50 + 30 + 20) = 100$$

Interpretation. This is an undiscounted cash-flow total, not the market value of the swap.

Check your reasoning. No principal is exchanged under this plain single-currency swap.

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5. Combine fixed debt and a swap - Answer B

Step 1. Write the original debt cost. The company still owes 6% to its lender.

Step 2. Add paid and subtract received. Final rate = 6% + SOFR - 6%.

Step 3. Cancel the fixed terms. The remaining annual financing rate is SOFR.

$$\text{Cost} = 6\% + SOFR - 6\% = SOFR$$

Interpretation. Falling SOFR now helps the company, but rising SOFR makes the combined cost higher.

Check your reasoning. The original bond contract was not rewritten; the swap changed the combined exposure.